

1. What is an Exception?

An **exception** is an unwanted or unexpected event that occurs during the execution of a program and interrupts the normal flow of the program.

Common types of exceptions in Java

- `ArithmeticException` – occurs when dividing by zero.
- `NullPointerException` – occurs when accessing a null object.
- `ArrayIndexOutOfBoundsException` – occurs when accessing an invalid array index.
- `NumberFormatException` – occurs when converting an invalid string into a number.

Program for Exception Handling:

```
public class ExceptionExample {  
    public static void main(String[] args) {  
  
        try {  
            int a = 10;  
            int b = 0;  
            int result = a / b;  
  
            System.out.println("Result: " + result);  
        }  
        catch (ArithmeticException e) {  
            System.out.println("Exception occurred: Cannot divide by zero");  
        }  
    }  
}
```

```
        System.out.println("Program continues...");
    }
}
```

O/P:

Exception occurred: Cannot divide by zero

Program continues...

2. What is Collections?

Collections in Java are a framework used to **store, manage, and manipulate a group of objects**.

Java provides classes such as:

- ArrayList
- LinkedList
- HashSet
- HashMap
- TreeSet

To remove duplicate names, we can use **HashSet**, because a **HashSet** does not allow duplicate elements.

Program to Remove Duplicate Names Using Collections:

```
import java.util.*;
```

```
public class RemoveDuplicates {
    public static void main(String[] args) {
```

```
ArrayList<String> names = new ArrayList<>();
```

```
names.add("Rekha");
```

```
names.add("Anu");
```

```
names.add("Priya");
```

```
names.add("Rekha");
```

```
names.add("Anu");
```

```
names.add("Divya");
```

```
System.out.println("Names before removing duplicates:");
```

```
System.out.println(names);
```

```
HashSet<String> uniqueNames = new HashSet<>(names);
```

```
System.out.println("Names after removing duplicates:");
```

```
System.out.println(uniqueNames);
```

```
}
```

```
}
```

O/P:

Names before removing duplicates:

[Rekha, Anu, Priya, Rekha, Anu, Divya]

Names after removing duplicates:

[Rekha, Anu, Priya, Divya]